

Quiz 4

CSE461: Introduction to Robotics

Department of Computer Science and Engineering

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- 1) What is the importance of weights in the law of machine learning models?
(2 marks)

Weights assigns importance to feature values.

Let say, $output = w_1 * feature_1 + w_2 * feature_2 + \dots + w_n * feature_n$

Here, in the law of model, we can notice that weights $w_1, w_2, \dots, w_n$ putting values to features. For an example, if $w_1 < w_2 < \dots < w_n$, we can understand that, (feature_n) is contributing more ~~to~~ than other features.

- 2) For the following input image I, if we apply a kernel on it, determine the feature matrix (Z). (Consider Stride = 1, no padding, Use max pooling(2x2)) (7 marks)

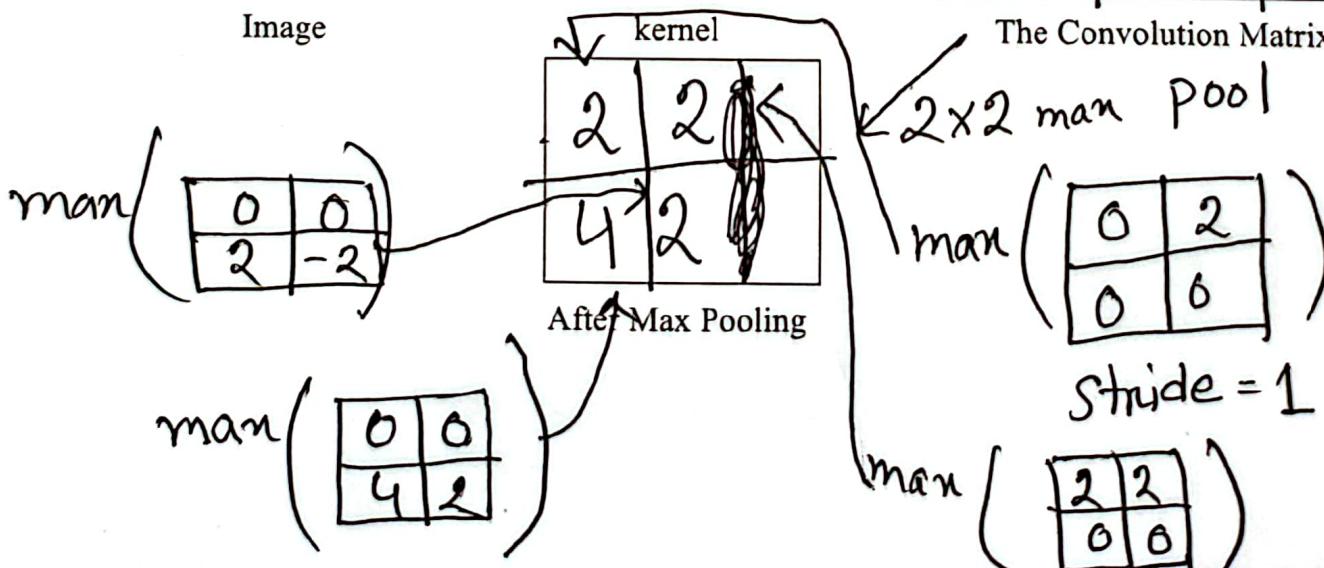
0	1	1	1	0
1	0	0	0	1
1	0	0	0	1
1	0	0	0	1
0	1	1	1	0

Image

1	1	0
0	-1	-2
-1	1	2

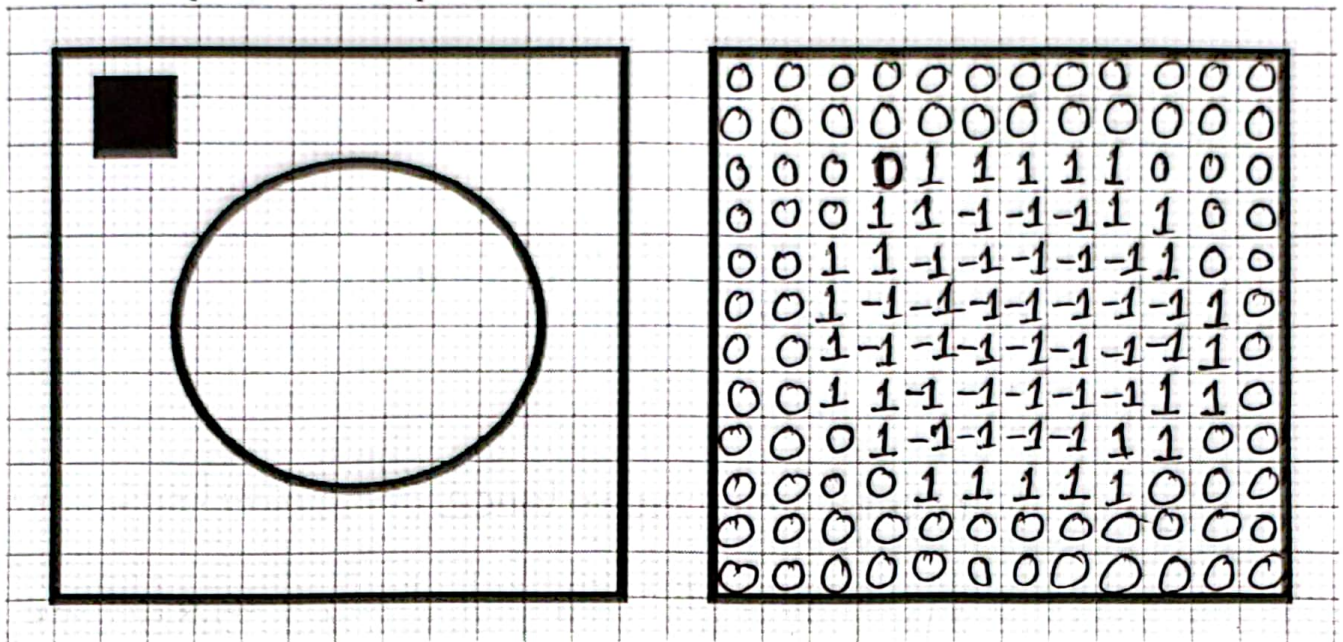
0	2	2
0	0	0
4	2	-2

The Convolution Matrix



3) What will be the final result of Frontier Based Mapping (3 marks)

For free space = 0, Occupied = 1, Unknown = -1



4) Share an interesting, unexpected, or memorable experience you have had at BRAC University so far. (3 marks)

None